

AOX target-engagement assay

Does SHAM or n-propyl gallate actually inhibit azide-resistant respiration in *Z. tritici* IPO323, at 15 and 27 °C, at the concentrations used in the growth experiment? One plate per temperature, 24 vials each, no additions after sealing.

Why this experiment

The growth experiment showed SHAM suppressing growth at 27 °C and not at 15 °C, while nPG did nothing at either. That difference has no explanation while target engagement is unmeasured. Here you block cytochrome c oxidase with azide, so whatever oxygen consumption remains must run through the alternative route, then ask whether each inhibitor removes it.

Result	What it means
nPG removes it at 0.1 mM	nPG engaged AOX at the dose you used. 'Underdosed' is dead, and the growth difference needs a different explanation.
nPG removes it at 1.0 mM but NOT at 0.1 mM	The cleanest possible outcome: nPG does inhibit AOX here, just not at the concentration your growth experiment reached. Underdosing becomes an evidenced explanation rather than an excuse, and you know the dose for any future run.
nPG removes it at neither dose	nPG does not engage AOX in this fungus at all. Its silence in the growth experiment says nothing about AOX, and you can state that.
SHAM removes it, nPG does not at either dose	Real differential engagement, and the explanation you have been looking for.
SHAM does NOT remove it	The SHAM growth effect at 27 °C is not acting through AOX. Uncomfortable, but decisive, and much better found now than by a referee.
No azide-resistant respiration is measurable	Inconclusive about the inhibitors, but informative: little alternative flux under these conditions.

STEP 0 — before you start

Use **OD600 = 0.005** in the vial — ten times your usual growth inoculum, and the practical ceiling before the optical sensors stop reading reliably. Spin down a culture and resuspend in fresh YMS. Prepare enough for 44 vials plus dead volume: about **250 mL**.

At this density, from the K values in your own control wells, expect roughly **0.3 mg/L/h** of oxygen consumption — about 2.5 mg/L over 8 h. That is ample for the model to resolve, but it means this is **not** a 90-minute acute assay. Run it for 6–12 h, or simply leave it overnight as you would any other plate.

Reader calibrated as usual. A vial of plain air-shaken YMS should read ~7–8 mg/L.

STEP 1 — sodium azide stocks

Azide goes in as **10 µL** per vial, dissolved in **water**. The concentration that fully blocks cytochrome oxidase in *Z. tritici* is not established in the literature, which is why the plate contains a titration rather than a single guess.

Want in vial	Make stock	Weigh NaN ₃	Dissolve to	Use
0.5 mM	250 mM	16.3 mg	1.00 mL water	titration
2 mM	1 M	65.0 mg	1.00 mL water	working conc.
5 mM	2.5 M	162.5 mg	1.00 mL water	titration

NaN₃ M_r = 65.01. General rule: mg = mM(stock) × 0.065 per mL. Sodium azide is very soluble (~40 g/100 mL), so all three go into solution easily.

STEP 2 — inhibitor stocks

Inhibitors go in as **20 µL** per vial, in ethanol. Each compound is tested at the dose used in the growth experiment AND at 5–10× that dose, so the two compounds are treated symmetrically. Published work on intact fungal cells commonly uses SHAM at 1 mM, five times your growth-experiment dose, so the high SHAM arm matters as much as the high nPG one.

Want in vial	Stock	How to make 1 mL
SHAM 0.2 mM	50 mM in EtOH	200 µL of your 250 mM master + 800 µL EtOH
SHAM 1.0 mM	250 mM in EtOH	your existing master, used neat — nothing to weigh
nPG 0.1 mM	25 mM in EtOH	100 µL of the 250 mM below + 900 µL EtOH
nPG 1.0 mM	250 mM in EtOH	53.1 mg n-propyl gallate in 1.00 mL absolute ethanol (M _r 212.2); vortex until clear

Ethanol works out at 0.40% v/v in every vial, identical across all 24. That is twice the growth experiment's 0.20%, which is fine because it is constant within this experiment — but keep it in mind if you ever compare absolute rates between the two experiments.

STEP 3 — fill (every vial gets 10 µL azide + 20 µL inhibitor)

Each vial: **5.03 mL cell suspension**, then **10 µL from the azide slot**, then **20 µL from the inhibitor slot**. Vials with no azide get 10 µL plain water; vials with no inhibitor get 20 µL plain ethanol. Water and ethanol are then identical in all 24.

n	azide slot (10 µL)	inhibitor slot (20 µL)	final in vial	what it tells you
3	water	ethanol	—	total respiration
1	250 mM NaN ₃	ethanol	azide 0.5 mM	titration: is cyt ox blocked yet?
3	1 M NaN ₃	ethanol	azide 2 mM	AZIDE-RESISTANT RESPIRATION (reference)
1	2.5 M NaN ₃	ethanol	azide 5 mM	titration: has it plateaued?
3	1 M NaN ₃	SHAM 50 mM	azide + SHAM 0.2 mM	SHAM at the dose you used
3	1 M NaN ₃	SHAM 250 mM	azide + SHAM 1.0 mM	SHAM at 5x (the usual literature dose)
3	1 M NaN ₃	nPG 25 mM	azide + nPG 0.1 mM	nPG at the dose you used
3	1 M NaN ₃	nPG 250 mM	azide + nPG 1.0 mM	nPG at 10x
1	water	SHAM 250 mM	SHAM 1.0 mM	high SHAM alone, no azide
1	water	nPG 250 mM	nPG 1.0 mM	high nPG alone, no azide
2	water	ethanol	no cells	cell-free blank

24 vials, OD600 0.005 in every cell-containing vial. Fill both plates from the same cell batch, alternating between them and swirling between draws, so 15 °C and 27 °C get identical cells.

STEP 4 — run and read

Seal, load, log for 6–12 h (overnight is fine). The cells will grow a little during this, so a raw straight-line slope would underestimate the starting rate. Instead fit the model you already use,

$$O(t) = O_{2,0} + (K/r)(1 - e^{-rt})$$

and read **K**, the oxygen consumption rate at the start of the window. K is the quantity this experiment is about: respiration under each treatment, before growth differences accumulate. Azide will also slow growth, which changes *r* — that does not matter here, because K is estimated at the window start under each treatment separately.

Use the same fitting-interval rule for every vial. Given how much this project has already been bitten by inconsistent windows, use a fixed rule rather than hand-selecting.

STEP 5 — the calculation

Quantity	How
Cytochrome-blocked?	K should FALL from 'total' to 'azide 2 mM' and then flatten between 2 and 5 mM. If 5 mM is much lower than 2 mM, cytochrome oxidase was not fully blocked at 2 mM — use 5 mM as your reference instead.
Azide-resistant respiration	K(azide 2 mM) – K(blank)
SHAM-sensitive fraction	K(azide) – K(azide + SHAM 0.2), and separately at 1.0 mM, each as a % of azide-resistant respiration
nPG-sensitive fraction	K(azide) – K(azide + nPG 0.1), and separately at 1.0 mM. Comparing 1x with 5–10x for BOTH compounds is the dose-engagement answer.
Compare temperatures	Both of the above at 15 vs 27 °C. Your hypothesis predicts a larger azide-resistant fraction at 27 °C.

Safety — sodium azide

Acutely toxic by ingestion, inhalation and skin contact. Wear gloves; weigh in a fume hood. **Never acidify** — it releases hydrazoic acid, which is toxic and explosive. **Never pour down the sink** — it forms shock-sensitive metal azides in copper and lead plumbing. Collect all azide-containing waste, including the vials after the run, in a labelled aqueous waste container. Azide also inhibits your own cytochrome oxidase; treat spills seriously.

What this does and does not settle

It measures whether your concentrations engage the alternative respiratory route. It does not prove that route is AOX — azide-resistant respiration can include other oxidases and non-mitochondrial oxygen use, which is why the cell-free blank matters. And forcing flux through the alternative route with azide shows capacity, not the contribution that route makes during untreated growth. Read it as a discrimination experiment, not a proof.